

H A C K A T H O N

Closing the Last Mile: Bridging the R3 Accuracy Gap

Problem Statement and Participant Guide

Duration: 1 Weeks | Team Size: Up 2 to 4 | HiLabs | Provider Data Accuracy

1. Context

R3 is HiLabs' provider attribute validation engine and the backbone of the PDA product line. It ingests provider records, runs them through a multi-layered evidence pipeline (web scraping, LLM parsing, domain trust scoring, ML classification), and outputs a validation label: Accurate, Inaccurate, or Inconclusive.

Today, R3's overall accuracy stands at approximately 75%. This is validated against a ground truth established by our Calling QC team, who verify provider addresses through direct phone outreach to practices, facilities, and offices.

The remaining ~25% gap represents records where R3's validation label disagrees with the Calling QC verdict. The Web validation accuracy is 88% for address attribute. In other words, the internet agrees with R3, but the phone call tells a different story.

Previous attempts to close this gap through additional data sources (claims data via FHIR, supplementary registries) have yielded only 1-1.5% net accuracy improvement. This hackathon asks you to think differently about the problem.

2. Problem Definition

2.1 What You Are Solving

You will be given a labelled dataset of 1,500 provider records that have been validated by both web research and Calling QC (human ground truth). A subset of these records (~400) show disagreement between R3 and Calling QC. Your task is to improve R3's effective accuracy on this dataset without degrading performance on records where R3 is currently correct.

All datasets required to work on this problem will be provided to participating teams at kickoff. No external data acquisition is needed to get started.

2.2 The Constraint

Do not break what already works. R3's current recommendations on the ~80% agreement zone are trusted by clients and operationally relied upon. Any solution you propose must preserve accuracy on this segment. The hackathon evaluates net accuracy, not just improvement on the gap.

2.3 R3 (Web) & Robocalling Budget

The R3 cost is \$35 per 1000 (3.5 cents/record) records while the robocalling cost is 50 cents/successful call record. This makes robocalling a constrained resource:

Parameter	Value
Allocation Limit	Max 450 records (30% of 1,500)
Conclusive Rate	40% of robocalled records return a conclusive Accurate / Inaccurate result. The remaining 60% return Inconclusive (no answer, voicemail, or ambiguous response).
Effective Yield	~180 usable verdicts if you use the full 450-call budget.

Robocalling alone cannot close the entire gap. The question is: which records do you choose to call, and what do you do about the rest?

3. Hackathon Tracks

Your team's submission must cover at least two of the three tracks below. Track 1 is mandatory. You may choose either or both of Tracks 2 and 3.

Track 1: Problem Discovery and Pattern Analysis (Mandatory)

Objective: Systematically analyze the ~400 disagreement records and identify structural patterns that explain why R3 (and the web) disagree with Calling QC.

Key Questions to Investigate (non-exhaustive):

- Are certain specialties or provider types disproportionately represented in the disagreement set?
- Do specific geographies or specialties show higher misalignment rates?
- Is there a time-decay pattern? Do recently updated records disagree more or less than stale ones?
- What does the claims data tell you about these providers? Are they actively billing from the address R3 validated, or from a different one?
- Are there identifiable failure archetypes such as providers who moved recently, group practices with multiple locations, telehealth-primary providers with virtual addresses, or shared medical spaces?

Deliverable: A structured analysis with a proposed segmentation taxonomy of disagreement patterns, supported by data. Each segment should suggest a different resolution strategy.

Track 2: Signal Engineering and Passive Accuracy Improvement

Objective: Engineer new signals or combine existing ones to resolve some portion of the disagreement records without active outreach.

Direction Ideas (non-exhaustive):

- **Claims-based address corroboration:** Can you match provider NPI + billing address in claims data to validate or contradict R3's output? Does recency of claims activity at a given address correlate with Calling QC's verdict?
- **Multi-source triangulation:** Can you combine signals from the NPI registry, state licensing boards, DEA registrations, hospital affiliations, or scheduling platforms (Zocdoc, Healthgrades, etc.) to build a confidence score that predicts when web-based evidence is likely wrong?
- **Temporal signals:** Does the gap between the last known web update and the calling date predict disagreement? Can you build a staleness score for web evidence?
- **NLP on Call QC comments:** The Call QC data includes free-text notes from calling agents. Can you extract structured insights that reveal systematic evidence gaps?

Deliverable: A working prototype or detailed methodology showing how your proposed signals would reclassify disagreement records, with measured precision and recall against the Calling QC ground truth.

Track 3: Smart Robocalling Allocation and Triage

Objective: Design an intelligent triage system that decides which records to send to robocalling in order to maximize accuracy lift under the budget constraints.

Given that you can only call 450 records and only ~180 will yield conclusive results, your triage model needs to maximize two things simultaneously:

1. **Likelihood of disagreement:** Prioritize records where R3's output is most likely to be wrong.
2. **Likelihood of conclusive call outcome:** Some records are inherently harder to verify by phone (e.g., large hospital systems with complex IVRs, solo practitioners who never answer). Prioritize records where a call is likely to produce a clear answer.

Deliverable: A scoring model or decision framework that rank-orders the 1,500 records for robocalling priority, along with a simulation showing the expected accuracy lift.

4. Datasets

All the following datasets will be provided to participating teams at kickoff. You do not need to source any data independently.

4.1 Primary Dataset (1,500 Records)

Field	Description
Provider NPI	National Provider Identifier
Provider Name	Individual or organizational name
Specialty	Primary specialty classification
Address (Street, City, State, ZIP)	The address under validation
R3 Label	R3 output: Accurate / Inaccurate / Inconclusive
R3 Confidence Score	Model confidence (0 to 1)
Web QC Label	Manual web research verdict
Call QC Label	Phone verification verdict (ground truth)
Call QC Comments	Free-text notes from the calling agent
R3 Evidence Sources	Domains and registries R3 used to reach its verdict
Record Metadata	State, region, provider type (individual / org), practice setting

4.2 Claims Data Supplement

Medical claims data for the same 1,500 records, including:

- Billing NPI and billing address from recent claims
- Service dates and claim recency
- Rendering vs. billing provider information
- Place of service codes

4.3 Supporting Documentation

- R3 architecture overview: evidence layers, signal weights, scoring methodology
- Robocalling platform overview: call flow, response taxonomy, conclusivity definitions
- Full data dictionary for all provided fields

5. Ground Rules

1	Preserve the agreement zone. Any changes to R3 labelling must not degrade accuracy on records where R3 and Calling QC currently agree. Net accuracy across all 1,500 records is what matters.
2	Robocalling budget is hard-capped. You may allocate a maximum of 450 records. You may choose to allocate fewer.
3	Conclusivity is fixed at 40%. You cannot assume a higher conclusivity rate. Model that 60% of robocalled records return Inconclusive.
4	No manual calling. All active verification must go through the robocalling channel only.
5	Internet research is allowed. You may use any publicly available sources, APIs, or datasets to engineer new signals. Document all external sources used.
6	Data confidentiality. This dataset is HIPAA-adjacent. Do not share it outside your team or upload it to any public service. Anonymize before any external API calls.

6. Evaluation

Evaluation happens on two layers:

Layer 1 – Train-Test Set (1,500 records, provided to you).

You are free to train, test, and tune your model on this dataset in any way you choose. The split strategy is up to your team.

Layer 2 - Validation Set (held by the organizing team, unseen to participants).

Your submitted code will be run against a separate holdout dataset that you will not have access to during the hackathon. Final accuracy scores and rankings are determined on this unseen set. Solutions that overfit to the 1,500 records will not transfer.

Submissions are scored on a 100-point composite rubric:

Dimension	Points	What We Are Looking For
Validation Set Accuracy	35	Net accuracy improvement on the unseen holdout. Deductions for degradation in the agreement zone. This is the primary signal.
Problem Discovery Depth	20	Rigor of the pattern analysis on the disagreement set. Quality and actionability of the segmentation taxonomy. Statistical support for claims.
Cost Efficiency	20	Judicious use/allocation expensive resources like robocalling. Cost/Run efficiency on R3, robocalling or other resources with a working solution (without current accuracy drop)

Feasibility and Scalability	15	Can this be productionized? Does it generalize beyond the 1,500-record sample? Are the data sources sustainable and automatable?
Presentation and Clarity	10	Quality of the final presentation. Clear articulation of approach, findings, and recommendations.

7. Submission Requirements

Each team must submit the following by the deadline:

- **Presentation Deck** covering your approach, findings, accuracy results on the training set, and your simulation of holdout performance. 15 minutes total (10 min presentation + 5 min Q&A). Teams of 2–4 participants. Every member should be able to speak to any part of the solution.
- **Code and Results** including all notebooks or scripts used, a README with setup instructions, training set results, and the output file your code would generate on the holdout dataset. The organizing team must be able to re-run your pipeline on the unseen validation set. Submission must include code repo, README with setup instructions, and a brief data quality report
- You need to publish the docker image with the required endpoints to query the data by assuming the write-only role created by the IT team. Each individual team will have a distinct role which will be assigned when the team is registered.
- Any language, framework, or cloud platform is permitted.
- Use publicly available data only (If needed). No proprietary payer datasets (except McKesson Compile).